(venv)projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

```
  GNU nano 8.6                                                    monitor.py
from watchdog.observers import Observer
from watchdog.events import FileSystemEventHandler
import time
import os

from hash_utils import calculate_hash
from logger import log_event
from config import SENSITIVE_DIRECTORIES

class FileMonitor(FileSystemEventHandler):

    def on_modified(self, event):
        if not event.is_directory:
            file_hash = calculate_hash(event.src_path)
            log_event(f"MODIFIED: {event.src_path} | HASH: {file_hash}")

    def on_created(self, event):
        if not event.is_directory:
            log_event(f"CREATED: {event.src_path}")
            self.check_sensitive(event.src_path)

    def on_deleted(self, event):
        log_event(f"DELETED: {event.src_path}")

    def on_moved(self, event):
        log_event(f"MOVED: {event.src_path} → {event.dest_path}")
        self.check_sensitive(event.dest_path)

    def check_sensitive(self, path):
        for sensitive in SENSITIVE_DIRECTORIES:
            if path.startswith(sensitive):
                log_event(f"ALERT: Sensitive file accessed → {path}")

if __name__ == "__main__":
    path = "./monitored_folder"
    event_handler = FileMonitor()
    observer = Observer()
    observer.schedule(event_handler, path, recursive=True)
    observer.start()
                                                [ Read 48 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was
```

projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

```
┌──(projectcsx㉿kali)-[~]
└─$ touch monitored_folder/test.txt
touch: cannot touch 'monitored_folder/test.txt': No such file or directory

┌──(projectcsx㉿kali)-[~]
└─$ touch monitored_folder/test.txt
touch: cannot touch 'monitored_folder/test.txt': No such file or directory

┌──(projectcsx㉿kali)-[~]
└─$

┌──(projectcsx㉿kali)-[~]
└─$ cd secure-file-monitor

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ touch monitored_folder/test.txt

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ echo "hello" >> monitored_folder/test.txt

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ mv monitored_folder/test.txt monitored_folder/sensitive/

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ touch monitored_folder/test2.txt

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ mv monitored_folder/test2.txt monitored_folder/sensitive

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ touch monitored_folder/test3.txt

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ mv monitored_folder/test3.txt monitored_folder/sensitive

┌──(projectcsx㉿kali)-[~/secure-file-monitor]
└─$
```

(venv)projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

```
  GNU nano 8.6                                                    logger.py
import logging
from config import LOG_FILE

logging.basicConfig(
    filename=LOG_FILE,
    level=logging.INFO,
    format="%(asctime)s - %(message)s"
)

def log_event(message):
    logging.info(message)
```

[ Read 11 lines ]

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ^G Help | ^O Write Out | ^F Where Is | ^K Cut | ^T Execute | ^C Location | M-U Undo | M-A Set Mark | M-] To Bracket |
| ^X Exit | ^R Read File | ^\ Replace | ^U Paste | ^J Justify | ^/ Go To Line | M-E Redo | M-6 Copy | ^B Where Was |

(venv)projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

```
  GNU nano 8.6                                                    config.py
SENSITIVE_DIRECTORIES = [
    "./monitored_folder/sensitive"
]
LOG_FILE = "logs/file_activity.log"
```

[ Read 4 lines ]

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ^G Help | ^O Write Out | ^F Where Is | ^K Cut | ^T Execute | ^C Location | M-U Undo | M-A Set Mark | M-] To Bracket |
| ^X Exit | ^R Read File | ^\ Replace | ^U Paste | ^J Justify | ^/ Go To Line | M-E Redo | M-6 Copy | ^B Where Was |

(venv)projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

GNU nano 8.6 monitor.py

```
    def on_modified(self, event):
        if not event.is_directory:
            file_hash = calculate_hash(event.src_path)
            log_event(f"MODIFIED: {event.src_path} | HASH: {file_hash}")

    def on_created(self, event):
        if not event.is_directory:
            log_event(f"CREATED: {event.src_path}")
            self.check_sensitive(event.src_path)

    def on_deleted(self, event):
        log_event(f"DELETED: {event.src_path}")

    def on_moved(self, event):
        log_event(f"MOVED: {event.src_path} → {event.dest_path}")
        self.check_sensitive(event.dest_path)

    def check_sensitive(self, path):
        for sensitive in SENSITIVE_DIRECTORIES:
            if path.startswith(sensitive):
                log_event(f"ALERT: Sensitive file accessed → {path}")

if __name__ == "__main__":
    path = "./monitored_folder"
    event_handler = FileMonitor()
    observer = Observer()
    observer.schedule(event_handler, path, recursive=True)
    observer.start()

    print("Monitoring started... Press CTRL+C to stop")

    try:
        while True:
            time.sleep(1)
    except KeyboardInterrupt:
        observer.stop()
    observer.join()
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was

Player

Debian 11.x 64-bit

1 2 3 4 3:38

(venv)projectcsx@kali: ~/secure-file-monitor

Session Actions Edit View Help

```
2026-02-15 03:18:51,210 - CREATED: ./monitored_folder/test.txt
2026-02-15 03:18:51,214 - MODIFIED: ./monitored_folder/test.txt | HASH: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
2026-02-15 03:19:26,437 - MODIFIED: ./monitored_folder/test.txt | HASH: 5891b5b522d5df086d0ff0b110fbd9d21bb4fc7163af34d08286a2e846f6be03
2026-02-15 03:19:47,016 - MOVED: ./monitored_folder/test.txt → ./monitored_folder/sensitive/test.txt
2026-02-15 03:26:11,902 - CREATED: ./monitored_folder/test2.txt
2026-02-15 03:26:11,903 - MODIFIED: ./monitored_folder/test2.txt | HASH: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
2026-02-15 03:26:54,377 - MOVED: ./monitored_folder/test2.txt → ./monitored_folder/sensitive/test2.txt
2026-02-15 03:26:54,377 - ALERT: Sensitive file accessed → ./monitored_folder/sensitive/test2.txt

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ nano config.py

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ nano hash_utils.py

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ nano logger.py

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ nano monitor.py

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ python monitor.py
Monitoring started... Press CTRL+C to stop
^C

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$ cat logs/file_activity.log
2026-02-15 03:18:51,210 - CREATED: ./monitored_folder/test.txt
2026-02-15 03:18:51,214 - MODIFIED: ./monitored_folder/test.txt | HASH: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
2026-02-15 03:19:26,437 - MODIFIED: ./monitored_folder/test.txt | HASH: 5891b5b522d5df086d0ff0b110fbd9d21bb4fc7163af34d08286a2e846f6be03
2026-02-15 03:19:47,016 - MOVED: ./monitored_folder/test.txt → ./monitored_folder/sensitive/test.txt
2026-02-15 03:26:11,902 - CREATED: ./monitored_folder/test2.txt
2026-02-15 03:26:11,903 - MODIFIED: ./monitored_folder/test2.txt | HASH: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
2026-02-15 03:26:54,377 - MOVED: ./monitored_folder/test2.txt → ./monitored_folder/sensitive/test2.txt
2026-02-15 03:26:54,377 - ALERT: Sensitive file accessed → ./monitored_folder/sensitive/test2.txt
2026-02-15 03:31:58,499 - CREATED: ./monitored_folder/test3.txt
2026-02-15 03:31:58,500 - MODIFIED: ./monitored_folder/test3.txt | HASH: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
2026-02-15 03:32:12,985 - MOVED: ./monitored_folder/test3.txt → ./monitored_folder/sensitive/test3.txt
2026-02-15 03:32:12,985 - ALERT: Sensitive file accessed → ./monitored_folder/sensitive/test3.txt

┌──(venv)─(projectcsx㉿kali)-[~/secure-file-monitor]
└─$
```